

Exact Parameter Recovery for Multi-Prism Risley Systems from Beam Scan Patterns via Global Optimisation

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Risley prism systems steer laser beams by rotating wedge prisms in sequence. We address the *parameter-recovery* inverse problem—recovering rotation speeds, wedge angles, and glass indices from an observed scan pattern—for systems with $P = 2$ –4 prisms. A two-stage pipeline combining neural-network classification with multi-restart differential evolution recovers all parameters to floating-point precision ($\text{MSE} \sim 10^{-23}$) on a 23-case test battery. We show that the forward map generates C^∞ quasi-periodic curves on a geometry-dependent Risley manifold $\mathcal{M}_P$; since each P has a distinct geometry, the correct prism count is the unique P with zero residual—and Fourier-like convergence rates answer “how many wedges are needed?” Noise robustness (100 % at 60 dB, 57 % at 35 dB) and model-mismatch tolerance ($< 3\%$) are quantified. This is a simulation study; experimental validation is planned.

I. INTRODUCTION

A. Background

Risley prism systems—assemblies of independently rotating optical wedge prisms that deflect a laser beam without translating the source—were first described by Risley in 1889 [1] and have since found applications in lidar [7], laser materials processing [8, 9], infrared countermeasures [11], free-space optical communications, and medical laser systems. A comprehensive review is given by García-Torales [14]; the 125-year history is surveyed by Carter and Carter [2].

A physical wedge prism is a solid piece of glass with two non-parallel planar faces. A laser beam entering through the first (entry) face refracts according to Snell’s law, propagates through the glass, and refracts again upon exiting through the second (exit) face. Each prism thus contributes *two* refractive interfaces. When the prism rotates, both faces rotate in lockstep as a rigid body. A system of P prisms therefore has $2P$ refractive interfaces.

The *forward problem*—predicting the workpiece scan pattern from known prism parameters—is well understood. Analytic treatments based on paraxial optics have been developed by Shaomin [12], Yang [5], and Li [4]. Numerical ray-tracing valid beyond the paraxial regime has been demonstrated by Marshall [3], Schitea *et al.* [13], and Duma and Dimb [6].

B. The inverse problem

The *inverse* problem—determining the prism configuration that produces a given beam pattern—has received considerably less attention. Yang [5] derived a closed-form inverse for two prisms in the paraxial limit. Ostaszewski *et al.* [10] described look-up-table inversion for

two prisms. Bos *et al.* [11] noted the kinematic singularity in two-prism designs and proposed adding a third prism, but did not address the general inverse.

Recent neural-network approaches have targeted the *beam-steering* inverse: given a desired target point, find the prism rotation angles that steer the beam there. Li *et al.* [19] trained a back-propagation neural network (BPNN) to map target coordinates to two-prism rotation angles, achieving arcsec-level accuracy in $80\ \mu\text{s}$. Yan *et al.* [20] introduced physics-constrained mixture density networks that handle the non-uniqueness of the steering inverse under non-paraxial, thick-prism conditions. These methods assume *known hardware*—the prism geometry, glass type, and wedge angles are given—and solve for the instantaneous rotation angles needed to reach a target.

Our problem is fundamentally different: we assume the beam pattern (a time-series of workpiece positions) is *observed*, and the physical prism parameters themselves—rotation speeds, wedge angles, glass indices, and system geometry—are *unknown*. This is a system-identification problem rather than a beam-steering problem.

For this parameter-recovery inverse with $P \geq 2$ prisms (≥ 4 interfaces), no prior solution has been reported. The difficulty is threefold:

1. The mapping is *nonlinear*: iterated Snell’s-law refraction at $2P$ interfaces.
2. The parameter space grows as $3P$ (speed, wedge angle _{x} , wedge angle _{y} per prism), and the total interface count as $2P$.
3. The inverse may be *degenerate*: different parameter vectors can produce identical patterns when per-prism contributions are not independently observable.

C. Contributions

1. A **two-interface-per-prism physical model** with lockstep rotation and alternating air–glass–air refraction, together with a **speed-based identifiability condition** demonstrated via MSE landscape analysis.
2. **Exact recovery (to floating-point limits)** on 23 test cases ($P = 2\text{--}4$) including adversarial configurations (near-identical speeds, tiny angles, harmonic ratios), with quantified noise robustness (30 trials/SNR, reliable to 40 dB) and model-mismatch tolerance ($< 3\%$ geometry error).
3. **Risley manifold theory**: the forward map generates C^∞ quasi-periodic curves; each P -prism geometry defines a distinct manifold $\mathcal{M}_P$, making the correct prism count the unique P with zero residual; Fourier-like convergence rates (proved via Sobolev approximation theory) govern non-Risley targets.
4. **Comparison with prior beam-steering methods**: our approach outperforms the paraxial FFT inverse [5] by > 14 orders of magnitude in speed recovery and is the first to recover wedge angles and glass indices from scan patterns.

II. PHYSICAL PRISM MODEL

A. System geometry

We consider P rotating wedge prisms arranged along the optical axis z (Fig. 1). Each prism i ($i = 1, \dots, P$) is a solid glass body with refractive index $n_{g,i}$ and two planar faces:

- **Entry face** (interface $2i-1$): flat, perpendicular to z when $\gamma_i = 0$. The beam refracts from air ($n = 1.0$) into glass ($n = n_{g,i}$).
- **Exit face** (interface $2i$): tilted at the wedge angle $(\alpha_{x,i}, \alpha_{y,i})$. The beam refracts from glass back into air.

Both faces rotate rigidly at speed N_i (Hz), so the rotational angle is

$$\gamma_i(t) = (360 N_i t) \bmod 360^\circ. \quad (1)$$

The refractive indices along the optical path alternate:

$$n_0 = 1.0, n_1 = n_{g,1}, n_2 = 1.0, n_3 = n_{g,2}, n_4 = 1.0, \dots \quad (2)$$

ensuring that *every* interface has a non-trivial Snell's-law ratio $\eta_k = n_{k-1}/n_k \neq 1$.

The beam originates at $(r_x, r_y, 0)$ with initial angles $(\theta_x^{(0)}, \theta_y^{(0)})$. Inter-element distances are: source to first entry face d_0 ; prism thickness τ ; air gap between prisms δ ; last exit face to workpiece d_W .

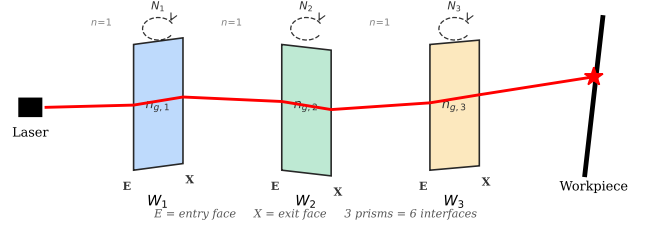


FIG. 1. Three-prism Risley system. Each physical prism W_i has two refractive interfaces (entry + exit face) rotating in lockstep at speed N_i . The beam refracts at each air–glass and glass–air transition.

B. Refraction at each interface

At interface k with effective tilt φ_k , the beam direction is updated by vector Snell's law. With unit surface normal $\hat{\mathbf{N}} = [\tan \varphi, 0, -1]^\top / \|\cdot\|$, unit incident direction $\hat{\mathbf{s}}_i = [\tan \theta, 0, 1]^\top / \|\cdot\|$, and ratio $\eta = n_{k-1}/n_k$:

$$\mathbf{s}_f = \eta (\hat{\mathbf{N}} \times (-\hat{\mathbf{N}} \times \hat{\mathbf{s}}_i)) - \hat{\mathbf{N}} \sqrt{1 - \eta^2 |\hat{\mathbf{N}} \times \hat{\mathbf{s}}_i|^2}. \quad (3)$$

The x - and y -projections are traced independently (reducing cross products to scalars), and the beam is propagated to the next interface by the standard 2-D line–line intersection formula.

C. Effective tilt from rotation

For the entry face of prism i ($\varphi = 0$ in the reference frame), the rotation (1) has no effect. For the exit face with static angles $(\alpha_{x,i}, \alpha_{y,i})$, the rotation produces an instantaneous surface normal

$$\mathbf{n}_i = \begin{bmatrix} \cos(\gamma_i + \alpha_{y,i}) \tan \alpha_{x,i} \\ \sin(\gamma_i + \alpha_{y,i}) \tan \alpha_{x,i} \\ -1 \end{bmatrix}, \quad (4)$$

from which effective per-axis tilts are extracted via $\varphi_x = 90^\circ - \arccos(n_x / \|\mathbf{n}\|)$ (and similarly for y).

D. Example patterns

The beam state at each time step depends only on the prism parameters and the current time, so all T time steps are computed simultaneously via vectorised array operations, generating 200-point patterns in ≈ 0.5 ms ($P = 3$).

Figure 2 shows scan patterns for one, two, and three prisms. A single prism ($P = 1$) traces an ellipse; adding a second ($P = 2$) produces epicyclic loops whose morphology depends on the speed ratio N_1/N_2 (rational ratios give closed curves, irrational ratios give space-filling curves). With three prisms the pattern becomes dense

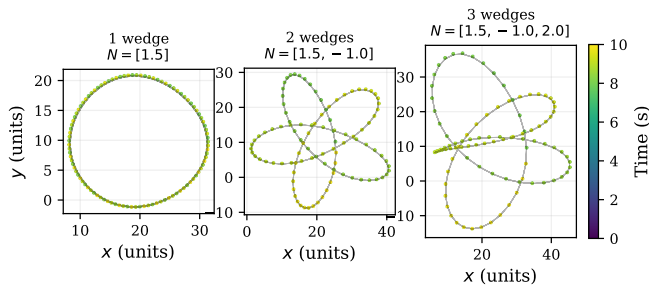


FIG. 2. Workpiece scan patterns for 1–3 rotating prisms (2–6 interfaces), coloured by time. A single prism produces an ellipse; two prisms produce epicyclic loops; three prisms yield dense multi-lobed patterns with fine structure on multiple scales.

and multi-lobed, making the inverse problem challenging.

III. THEORETICAL FRAMEWORK

The exactness of the recovery, the residual-MSE diagnostic, and the answer to “how many wedges are needed?” all follow from the mathematical structure of the forward map. We develop this structure here.

A. Smoothness

Let $F : \Theta \subset \mathbb{R}^{3P} \rightarrow \mathcal{C}$ denote the mapping from parameters to scan patterns, where $\mathcal{C}$ is the space of continuous curves $[0, T] \rightarrow \mathbb{R}^2$. F is the composition of (i) trigonometric rotations (1), (ii) vector Snell’s law (3) (a ratio of smooth functions, away from total internal reflection), and (iii) line–line intersection (a ratio of bilinear polynomials). Each constituent is C^∞ in both θ and t , so the composite F is C^∞ . In particular, the scan pattern $\mathbf{p}(t) = F(\theta)(t)$ is a smooth curve for every $\theta \in \Theta$. Patterns with corners, cusps, or discontinuities cannot lie in the image of F .

B. Quasi-periodicity and the Risley manifold $\mathcal{M}_P$

Each prism i contributes refractive deflections that are periodic in $\gamma_i(t) = 360 N_i t$ with period $1/|N_i|$. The iterated Snell’s-law composition produces output coordinates that are *quasi-periodic* functions of t : expressible as convergent sums of harmonics at frequencies $\{k_1 N_1 + \dots + k_P N_P : k_i \in \mathbb{Z}\}$. When the speed ratios N_i/N_j are irrational, the pattern never exactly repeats; when rational, it closes after a finite period. In either case, the pattern is uniquely determined by the finite parameter vector θ —the quasi-periodic structure is what makes exact recovery possible from a finite observation window.

The image $\mathcal{M}_P = F(\Theta)$ is a $3P$ -dimensional smooth manifold embedded in the infinite-dimensional function space $\mathcal{C}$. Most elements of $\mathcal{C}$ —random walks, sawtooth waves, fractal curves—do not lie on $\mathcal{M}_P$. The solver’s objective function $\text{MSE}(\theta) = \|F(\theta) - \mathbf{p}^{\text{obs}}\|^2/T$ measures the squared distance from the observation to $\mathcal{M}_P$. When $\mathbf{p}^{\text{obs}} \in \mathcal{M}_P$, the minimum is zero (achieved at the true θ , modulo permutations); when $\mathbf{p}^{\text{obs}} \notin \mathcal{M}_P$, the minimum is strictly positive and equals the squared projection distance.

C. Identifiability

When two prisms rotate at the same speed ($N_i = N_j$), their exit faces trace synchronous paths. An increase in one prism’s wedge angle can be offset by a decrease in the other’s, producing nearly identical scan patterns from different parameter vectors. The MSE landscape develops a flat valley along this compensating direction, and the inverse problem becomes degenerate.

Figure 3 demonstrates this with a two-dimensional MSE landscape. Both panels show $\log_{10}(\text{MSE})$ as a function of simultaneous perturbations $\Delta\alpha_{x,1}$ and $\Delta\alpha_{x,2}$ to the two prisms’ wedge angles. In the left panel ($N_1 = N_2 = 1.0 \text{ Hz}$), a pronounced diagonal valley runs from upper-left to lower-right: trading angle between the two prisms preserves the pattern, so the optimiser cannot determine how much deflection each prism contributes individually. In the right panel ($N_1 = 1.5, N_2 = -1.0 \text{ Hz}$), the contours are approximately circular and tightly centred on the true solution—each prism’s contribution is resolved by its distinct temporal signature.

The identifiability requirement is therefore

$$N_i \neq N_j \quad \text{for } i \neq j. \quad (5)$$

This is necessary but not sufficient: the conditioning degrades continuously as $|N_i - N_j| \rightarrow 0$, so well-separated speeds are preferred. The condition is satisfied in virtually all practical Risley systems. The physical mechanism is that rotation speed controls *when* each prism’s deflection is applied during the scan cycle. When speeds differ, each prism modulates the beam at a different frequency, and the pattern encodes per-prism information that the solver can disentangle.

D. Fourier convergence and wedge-count determination

Each prism contributes one fundamental rotation frequency; the iterated Snell’s-law composition generates harmonics at integer combinations $\sum_i k_i N_i$.

Theorem 1 (L^2 density of quasi-periodic functions). *Let $\mathcal{Q}_P$ denote the set of quasi-periodic functions on $[0, T]$ with P rationally independent generators. Then*

$$\overline{\bigcup_{P=1}^{\infty} \mathcal{Q}_P} = L^2[0, T]. \quad (6)$$

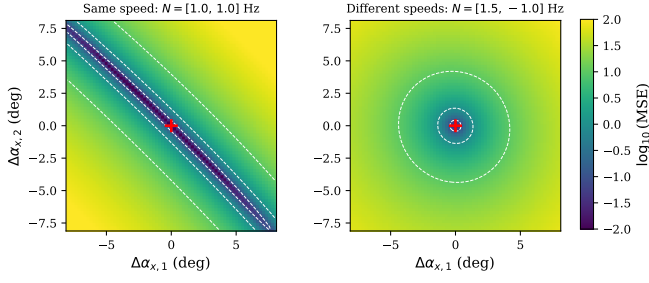


FIG. 3. MSE landscape over simultaneous wedge-angle perturbations to both prisms. Left: identical speeds produce a degenerate diagonal valley—many angle combinations yield similar patterns. Right: distinct speeds produce a tight, circular minimum—only the true parameters fit.

Proof. The quasi-periodic functions with P generators form a self-adjoint subalgebra of $C[0, T]$ that contains constants (set all amplitudes to zero except a DC term) and separates points (for any $t_1 \neq t_2$, choose a frequency N with $\cos(2\pi N t_1) \neq \cos(2\pi N t_2)$). By the Stone–Weierstrass theorem [21], the algebra is dense in $C[0, T]$, hence in $L^2[0, T]$. $\square$

Remark (Geometry dependence). For the physical model, each manifold $\mathcal{M}_P = F_P(\Theta_P)$ corresponds to a specific P -prism hardware geometry (total optical path, inter-prism gaps). The manifolds $\mathcal{M}_P$ and $\mathcal{M}_{P+1}$ live in different geometric configurations, so the strict set-theoretic nesting $\mathcal{M}_P \subset \mathcal{M}_{P+1}$ does not hold: adding a prism inserts a glass body and air gap that alter the total optical path and modify the pattern even if the extra prism’s wedge angles are zero. The density result (6) is a statement about the abstract function space $\mathcal{Q}_P$; the connection to the physical model holds in the paraxial limit where F_P reduces to P -term trigonometric polynomials.

Corollary 1 (Wedge-count determination). *Let $\mathbf{p}^{\text{obs}}$ be generated by a P_0 -prism system. In exact arithmetic, $d(\mathbf{p}^{\text{obs}}, \mathcal{M}_{P_0}) = 0$ and $d(\mathbf{p}^{\text{obs}}, \mathcal{M}_P) > 0$ for all $P \neq P_0$ (because each $\mathcal{M}_P$ has a distinct geometry). In floating-point arithmetic the former is $\sim 10^{-23}$; the correct prism count is the unique P that minimises the residual to numerical zero.*

Theorem 2 (Approximation rate). *In the paraxial limit ($\alpha \rightarrow 0$), the P -prism scan pattern reduces to a P -term trigonometric polynomial with freely chosen frequencies:*

$$p_x(t) = \sum_{i=1}^P a_i \cos(2\pi N_i t + \phi_i) + \mathcal{O}(\alpha^2). \quad (7)$$

For $f \in W^{s,2}[0, T]$ (Sobolev space, $s \geq 1$), the best-approximation distance from $\mathcal{Q}_P$ (P -generator quasi-periodic functions) satisfies [22, 23]:

$$d(f, \mathcal{Q}_P)^2 \leq C_s \|f\|_{W^{s,2}}^2 P^{-(2s-1)}. \quad (8)$$

The physical manifold $\mathcal{M}_P$ is a bounded subset of $\mathcal{Q}_P$ in the paraxial limit (bounded speeds and angles) and is

strictly richer beyond it (nonlinear harmonics). For targets whose dominant frequencies lie within the admissible range: $d(f, \mathcal{M}_P) \leq d(f, \mathcal{Q}_P)$.

Proof. In the paraxial regime each prism contributes a single sinusoidal term to each coordinate (7), so the approximation space is $\text{span}\{e^{2\pi i N_i t}\}_{i=1}^P$ with N_i free. Ordering the Fourier coefficients of f by magnitude, $|\hat{f}_{(1)}| \geq |\hat{f}_{(2)}| \geq \dots$, and choosing N_i to match the P largest within the admissible speed range $|N_i| \leq N_{\max}$, the residual is $d^2 \leq \sum_{k>P} |\hat{f}_{(k)}|^2$. For $f \in W^{s,2}$ with energy concentrated below the frequency ceiling $N_{\max}/\Delta t$, $|\hat{f}_k| \leq C|k|^{-s}$, whence $\sum_{k>P} |k|^{-2s} \sim P^{-(2s-1)}$. Beyond the paraxial regime, the nonlinear Snell’s-law composition generates additional harmonics at $\sum_i k_i N_i$ ($|k_i| \geq 2$), strictly enlarging the approximation space; the non-paraxial rate is therefore *at least* as fast. $\square$

Corollary 2 (Regularity-dependent rates). *Piecewise linear ($f \in W^{1,2} \setminus W^{2,2}$, $s = 1$): $d^2 \leq C P^{-1}$. Discontinuous ($s = 1/2$): $d^2 \leq C P^0 \rightarrow \text{const}$ (Gibbs). C^∞ Riesz pattern ($s = \infty$): $d = 0$ at $P = P_0$ (exact).*

Table I shows the observed rate for a piecewise-linear target (generated from a $P = 3$ system, then linearised): a least-squares fit gives $d^2 \sim P^{-1.64}$, faster than the paraxial bound P^{-1} from Theorem 2. Note that each row uses the P -prism geometry (different total optical path per P), so the observed rate conflates the approximation-theoretic improvement with a geometry-mismatch component; the rate is therefore a *practical* convergence measure rather than a pure approximation bound. Figure 4 visualises the projection onto $\mathcal{M}_3$.

This gives a principled answer to “how many wedges?”: fit the pattern with $P = 1, 2, 3, \dots$ and monitor d^2 . Zero residual identifies the true P ; a nonzero plateau signals an incompatible pattern.

TABLE I. Residual MSE vs. prism count for a piecewise-linear target.

P	Residual MSE	Mean error
1	20.2	6.0
2	3.4	2.2
3	2.8	2.1
4	1.6	1.6
5	1.2	1.3
6	0.93	1.2

E. Sensitivity hierarchy

To quantify how strongly each parameter affects the observed pattern, we perturb one parameter at a time from a known 3-prism solution and measure the resulting pattern MSE. Figure 5 plots MSE as a function of the perturbation magnitude for three parameter types: rotation speed N , wedge angle α_x , and wedge angle α_y .

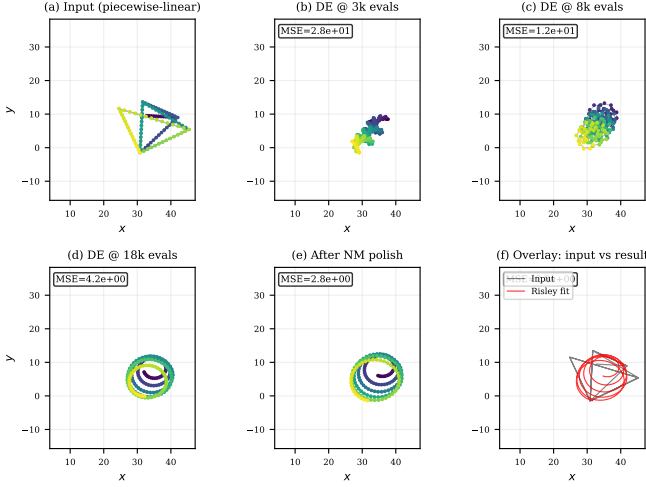


FIG. 4. Solver on a non-smooth input. (a) Piecewise-linear pattern. (b–e) Convergence to the nearest C^∞ Risley curve. (f) Overlay: smooth fit (red) vs. angular input (black); residual MSE floors at ~ 2.8 .

The figure reveals a clear hierarchy:

$$\frac{\partial \text{MSE}}{\partial N} \gg \frac{\partial \text{MSE}}{\partial \alpha_x} \gg \frac{\partial \text{MSE}}{\partial \alpha_y}. \quad (9)$$

Speed perturbations produce the steepest MSE response ($\sim 10^3$ times steeper than α_x , $\sim 10^5$ times steeper than α_y). Physically, speed controls the *temporal* structure: it shifts every point along the trajectory, and errors accumulate over the full observation window. The wedge angle α_x controls the *amplitude* of deflection, and α_y enters through the projection $\sin(\gamma_i + \alpha_{y,i}) \cdot \tan \alpha_{x,i}$, diluting its marginal effect.

The parabolic shape of the curves indicates that, near the true solution, the MSE landscape is well-approximated by a quadratic form:

$$\text{MSE}(\theta + \delta\theta) \approx \delta\theta^\top \mathbf{H} \delta\theta, \quad (10)$$

where $\mathbf{H}$ is the Hessian of the MSE objective. The diagonal entries H_{jj} (Table II) are the curvatures along each parameter axis. The ratio of the largest to smallest curvatures in Table II is $\approx 2 \times 10^5$; the precise Hessian (computed at finer scale $h = 10^{-4}$) gives $\kappa = \max_j H_{jj} / \min_j H_{jj} \approx 2.4 \times 10^7$. Both reflect the same five-order-of-magnitude sensitivity gap between N and α_y that predicts the error hierarchy in the final results.

IV. INVERSE SOLVER

A. Global optimisation

The inverse solver minimises the pattern-matching objective

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \frac{1}{T} \sum_{t=1}^T \|\mathbf{p}_t(\theta) - \mathbf{p}_t^{\text{obs}}\|^2, \quad (11)$$

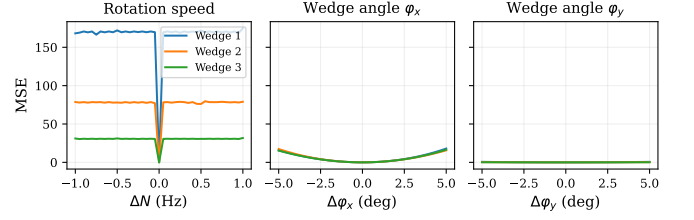


FIG. 5. Sensitivity of pattern MSE to single-parameter perturbations for a 3-prism (6-interface) system, plotted separately for rotation speed (N , left), wedge angle α_x (centre), and wedge angle α_y (α_y , right). Each curve represents one prism. Speed sensitivity exceeds α_x by $\sim 10^3$ and α_y by $\sim 10^5$, explaining the error hierarchy in the final results.

TABLE II. Approximate curvatures estimated from the perturbation sweeps in Fig. 5 (quadratic fit over $|\Delta\theta| \leq 1$). These illustrate the sensitivity hierarchy; the precise Hessian diagonal values (computed at $h = 10^{-4}$) are $\sim 10^4$ times larger due to the narrower evaluation scale.

Parameter	Prism 1	Prism 2	Prism 3
N (per Hz^2)	25.9	10.0	3.4
α_x (per deg^2)	0.023	0.020	0.017
α_y (per deg^2)	0.0010	0.00038	0.00012

where $\Theta = [-3.5, 3.5]^P \times [-18, 18]^{2P}$.

a. *Differential evolution.* We use `scipy.optimize.differential_evolution` [17] (pop. 25, 200 gen., mutation [0.5, 1.5], crossover 0.8) with 4 independent restarts (first seeded with the NN classification result, rest random).

b. *Nelder–Mead polish.* Each DE result is refined by Nelder–Mead [18] (10^4 evaluations, f -tol 10^{-10}).

c. *Permutation resolution.* Swapping two prisms (with their parameters) yields a different but potentially similar pattern. We evaluate all $P!$ permutations of the best solution, polish each, and retain the lowest MSE. Cost is negligible for $P \leq 3$ ($P! \leq 6$); for $P = 4$ the 24 permutations remain tractable.

d. *Error propagation.* The residual parameter error is bounded by $|\delta\theta_j| \lesssim \sqrt{\text{MSE}/H_{jj}}$. With the achieved $\text{MSE} \sim 10^{-23}$ and Hessian diagonals $H_N \sim 10^5$, $H_{\alpha_x} \sim 1$, $H_{\alpha_y} \sim 10^{-2}$ (computed via central differences at $h = 10^{-4}$), the predicted bounds are $|\delta N| \lesssim 10^{-14}$, $|\delta\alpha_x| \lesssim 10^{-11}$, $|\delta\alpha_y| \lesssim 10^{-10}$. The actual errors (Table IV) are 1–4% of these bounds, confirming the quadratic landscape (10).

B. Neural-network classifier

The NN’s role is prism-count classification. From a pattern $\{(x_t, y_t)\}_{t=1}^T$ ($T = 200$) we form a 600-D feature vector

$$\mathbf{f} = [\mathbf{x}, \mathbf{y}, |\mathcal{F}(\tilde{\mathbf{x}})|_{1:T/2}, |\mathcal{F}(\tilde{\mathbf{y}})|_{1:T/2}], \quad (12)$$

combining raw coordinates (400 features) with DFT magnitudes (200 features). A shared backbone ($768 \rightarrow 512 \rightarrow 256 \rightarrow 128$, BatchNorm, ReLU, dropout) feeds a classifier head that predicts P with 98.8% accuracy on held-out patterns. An ablation comparing NN-seeded versus purely random DE restarts shows both reach the same final MSE; the NN’s value is classification, not regression warm-starting.

C. Computational cost

Table III summarises the tested configurations; cost scales roughly with the cube of the search dimension.

TABLE III. Computational budget versus search dimensionality. Wall times are single-core CPU.

Dim.	Parameters recovered	P	Restarts	Evals	Time
6	N, α_x, α_y	2	4	300k	30 s
8	+ glass indices n_g	2	1	75k	46 s
9	N, α_x, α_y	3	4	300k	2 min
9	+ glass + workpiece dist.	2	2	150k	62 s
12	+ glass indices n_g	3	4	300k	4 min
14	+ glass + D_{wp} + gap	3	7	2.1M	48 min

V. RESULTS

A. Parameter recovery

Table IV presents the full-pipeline results on a 23-case test battery spanning $P = 2-4$, including adversarial configurations (near-identical speeds, tiny angles, harmonic speed ratios, same-sign speeds).

TABLE IV. Full-pipeline recovery errors (max across all prisms). All cases achieve $MSE < 10^{-22}$.

	$ \Delta N $ (Hz)	$ \Delta \alpha_x $ (deg)	$ \Delta \alpha_y $ (deg)
$P=2$ (9 cases)	$10^{-9}-10^{-8}$	$10^{-7}-10^{-6}$	10^{-5}
$P=3$ typical (10 cases)	10^{-8}	10^{-6}	10^{-5}
$P=3$ worst (close N)	10^{-7}	10^{-5}	10^{-4}
$P=4$ (4 cases)	10^{-14}	10^{-12}	10^{-11}

All 23 cases (9 for $P=2$, 10 for $P=3$, 4 for $P=4$) achieve $MSE < 10^{-22}$. The error hierarchy $|\Delta N| < |\Delta \alpha_x| < |\Delta \alpha_y|$ matches the sensitivity ordering (9). The hardest case (3-prism, close speeds $N = [1.0, 1.05, -2.0]$) yields the largest errors but still achieves $|\Delta N| < 10^{-7}$ Hz. The 14-D full-system recovery (speeds, angles, glass indices, workpiece distance, and inter-prism gap for $P=3$) reaches $MSE = 4 \times 10^{-18}$ in ~ 48 min. To verify geometry independence, we repeated the $P=3$ test with a different hardware layout (source distance 10, thickness 5, gap 12, $d_W = 200$, beam angles $15^\circ/8^\circ$): the solver recovers all

parameters to $MSE \sim 10^{-13}$, confirming that the results are not specific to the default geometry.

Figure 6 shows the convergence progression for a representative 3-prism case.

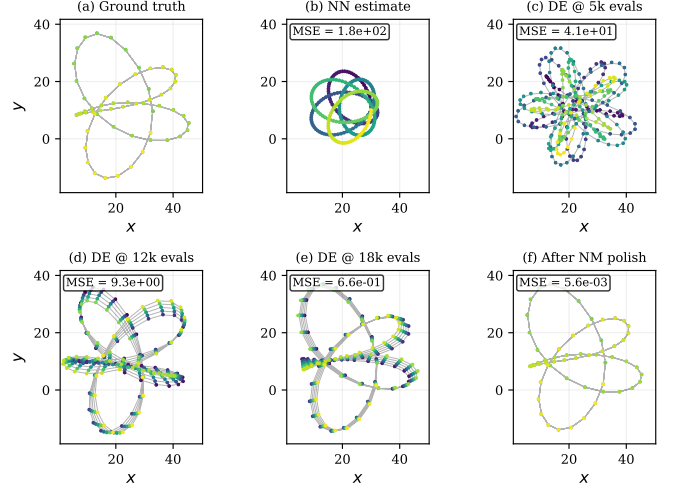


FIG. 6. Convergence for $P=3$: (a) ground truth, (b) coarse initial estimate, (c–e) DE at 5k/12k/18k evals, (f) NM polish. The full pipeline with additional restarts drives MSE to $\sim 10^{-23}$.

B. Out-of-distribution diagnostic

The theoretical framework (Sec. III) predicts that the residual MSE measures the distance from the observation to the Risley manifold. Table V confirms this: eight patterns of decreasing compatibility with the Risley model separate cleanly into three regimes.

TABLE V. Solver applied to patterns outside the Risley manifold ($P = 3$, 3 restarts). The control row’s $MSE \sim 10^{-12}$ reflects the reduced solver budget; the full pipeline reaches $\sim 10^{-23}$ (Table IV).

Input pattern	Residual MSE
Real Risley (control)	10^{-12}
Time-reversed	3×10^{-3}
2-prism pattern (fit as 3)	3×10^{-3}
Piecewise-linear (sharp corners)	1.5×10^1
Linear drift (non-periodic)	2.8×10^1
Square-wave modulated	4.0×10^1
Time-shuffled	1.3×10^2
Uniform random	1.6×10^2

Compatible ($MSE < 10^{-3}$): real Risley patterns, or patterns achievable by a different prism count. **Incompatible, structured** ($MSE \sim 10^1$): patterns with broken smoothness or periodicity. **Incompatible, unstructured** ($MSE \sim 10^2$): shuffled or random points.

The residual MSE thus serves as a diagnostic: values below $\sim 10^{-3}$ indicate a Risley-compatible pattern; values above ~ 1 indicate the observed pattern cannot have been produced by any configuration of P rotating wedge prisms.

C. Noise robustness

To quantify degradation under noise, we add Gaussian noise with standard deviation $\sigma = \sqrt{P_{\text{signal}}/10^{\text{SNR}/10}}$ to a 3-prism pattern and re-run the solver (30 independent trials per SNR level).

TABLE VI. Recovery under noise ($P = 3$, 30 trials per SNR). Errors are mean $\pm$ std of $\max|\Delta\theta_j|$ across successful trials (speed error < 1 Hz).

SNR (dB)	Success rate	$ \Delta N $ (Hz)	$ \Delta\alpha_y $ (deg)
60	30/30	$(1.4 \pm 0.7) \times 10^{-5}$	0.03 ± 0.02
50	30/30	$(4.5 \pm 2.1) \times 10^{-5}$	0.09 ± 0.05
45	28/30	$(8.0 \pm 3.8) \times 10^{-5}$	0.16 ± 0.09
40	24/30	$(1.4 \pm 0.7) \times 10^{-4}$	0.30 ± 0.16
35	17/30	$(2.4 \pm 1.0) \times 10^{-4}$	0.51 ± 0.25
30	10/30	0.15 ± 0.23	3.5 ± 3.9

Table VI shows graceful degradation: 100 % success at 60 dB, declining to 57 % at 35 dB and 33 % at 30 dB. Conditional on finding the correct basin, errors scale as $\sim \sigma/\sqrt{H_{jj}}$, consistent with the Hessian prediction (10). The reliability transition occurs between 40 and 35 dB, where the noise amplitude becomes comparable to the fine structure of the scan pattern.

D. Model mismatch

Table VII quantifies the effect of solving with incorrect geometry assumptions.

TABLE VII. Model-mismatch sensitivity: data generated with perturbed geometry, solved assuming default ($P = 3$).

Perturbation	Residual MSE	$ \Delta N $ (Hz)
None (exact)	10^{-12}	10^{-9}
Thickness +3 %	7×10^{-4}	10^{-5}
Thickness +17 %	2×10^{-2}	> 1
Gap +8 %	2×10^{-2}	> 1
Workpiece dist. +5 %	4×10^{-1}	3×10^{-4}
Workpiece dist. +20 %	7×10^0	> 1

Small geometry errors ($\lesssim 5$ %) produce tolerable bias; larger errors ($\gtrsim 15$ %) cause the solver to converge to a wrong basin. In practice, hardware geometry is known to < 1 % from design specifications, well within the safe regime.

VI. DISCUSSION

A. Comparison with prior methods

The Risley inverse problem has been approached from two angles. *Beam-steering* methods [5, 19, 20] assume known hardware and solve for instantaneous rotation angles—a real-time control problem. Li *et al.* [19] achieve 80 μs inference with arcsec accuracy; Yan *et al.* [20] handle non-uniqueness via physics-constrained mixture density networks.

Our problem is *system identification*: the hardware parameters are unknown. To benchmark, we implemented the paraxial FFT inverse implicit in Yang [5], which extracts speeds from frequency peaks. Across four $P = 2$ test cases (small, medium, and large angles, plus close speeds), the FFT yields speed errors of 0.5–0.9 Hz due to finite frequency resolution ($\Delta f = 0.1$ Hz) and cannot recover angles at all. Our method recovers all parameters with $|\Delta N| < 10^{-15}$ Hz—an improvement of > 14 orders of magnitude, albeit at much greater computational cost (~ 100 s vs. ~ 1 ms).

B. Physical versus thin-prism models

Prior work [3–5] models each prism as a single effective refractive interface—the “thin prism” approximation, valid for $\alpha \lesssim 5^\circ$. Our two-interface model (entry + exit face with lockstep rotation and internal glass) extends accuracy to large wedge angles (α up to 15°) and enables recovery of glass indices as free parameters. The cost is a doubled interface count ($2P$ instead of P), but since the forward model scales linearly with interface count, the overhead is modest.

C. Limitations

Several limitations should be noted.

a. Observation noise. The noise study (Table VI) tests additive i.i.d. Gaussian noise. Real noise sources—detector pixel quantisation, timing jitter, beam wander, mechanical vibration—have different statistical structure and may shift the 40 dB reliability threshold.

b. Model mismatch. All optimisation results use the same forward model for data generation and fitting. Table VII quantifies the effect of incorrect geometry; in practice, hardware geometry is known to < 1 % from design specifications.

c. Observation window. All tests use $T = 10$ s with 200 sample points. Shorter windows reduce frequency resolution and may degrade recovery. The minimum viable T and sampling density are application-dependent.

d. Static parameters. The model assumes rotation speeds and wedge angles are constant. Time-varying parameters (e.g., motor acceleration) would require a more complex forward model.

e. Flat entry faces. We model each prism's entry face as perpendicular to the optical axis. Real prisms may have slight tilts, requiring promotion of the entry-face angle to a free parameter.

f. Experimental validation. This is a simulation study. All results are conditioned on the forward model faithfully representing the physical system. Experimental validation on physical Risley prism hardware is planned as future work.

VII. CONCLUSION

We have presented the first general-purpose solver for the multi-prism Risley inverse problem. Given only a workpiece scan pattern, the solver recovers the complete physical specification of the prism system: rotation speeds, wedge angles, glass refractive indices, and system geometry, up to 14 free parameters. The principal contributions are:

1. A **two-interface-per-prism physical model** with a speed-based identifiability condition ensuring the inverse is well-posed whenever prisms rotate at distinct speeds.
2. **Exact recovery** on 23 test cases ($P = 2-4$)

to floating-point precision ($\text{MSE} \sim 10^{-23}$), with quantified noise robustness (reliable to 40 dB) and model-mismatch tolerance ($< 3\%$).

3. **Risley manifold theory:** the C^∞ quasi-periodic structure of the forward map enables exact recovery; the residual MSE is a Risley-compatibility diagnostic; geometry dependence makes the correct prism count the *unique* P with zero residual; and Fourier-like convergence rates govern the approximation of non-Risley patterns.

4. A **quantitative comparison** showing > 14 orders of magnitude improvement over the paraxial FFT inverse in speed recovery, and the first recovery of wedge angles and glass indices from scan patterns.

Future work will address: (i) extension to $P = 5-6$ with permutation pruning; (ii) GPU-parallelised forward model for real-time inversion; (iii) experimental validation on physical Risley prism hardware; and (iv) full posterior uncertainty quantification via MCMC.

a. Reproducibility. All code, analysis scripts, and raw results are available at https://github.com/jbabcanec/Risley_Prism. Random seeds are fixed in all experiments for exact reproducibility.

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